Spyder

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C:\Users\wamul\OneDrive\Desktop\Computer-Logic-and-Programming\untitled7.py

untitled7.py* | untitled8.py* | untitled9.py* | untitled10.py* | untitled11.py*

```python
# -*- coding: utf-8 -*-
"""
Created on Thu Mar 12 09:53:15 2026

@author: wamul
ALLow the user to enter a principal amount and interest rate. Repeatedly (need a loop to control
the program execution) compute the annual interest (principle x rate). Compute ending balance
to be principal (beginning balance + interest). Display year, beginning balance and ending
balance for each of the first 5 years. Display the accumulated interest for the 5 years. Note: the
new balance by year (this will be the principle for the following year
"""

principal = float(input("Enter principal amount: "))
rate = float(input("Enter interest rate: "))

total_interest = 0

print("\nYear        Beginning Balance        Ending Balance")

for year in range(1, 6):
    beginning_balance = principal
    interest = beginning_balance * rate
    ending_balance = beginning_balance + interest

    total_interest += interest

    print(f"{year:<10}${beginning_balance:,.2f}        ${ending_balance:,.2f}")

    principal = ending_balance  # update for next year

print(f"\nTotal interest earned: ${total_interest:,.2f}")
```

Console 1/A

```
In [22]: %runfile C:/Users/wamul/OneDrive/Desktop/Computer-
Logic-and-Programming/untitled7.py --wdir
Enter principal amount: 10000
Enter interest rate: 0.09

Year        Beginning Balance        Ending Balance
1         $10,000.00        $10,900.00
2         $10,900.00        $11,881.00
3         $11,881.00        $12,950.29
4         $12,950.29        $14,115.82
5         $14,115.82        $15,386.24

Total interest earned: $5,386.24

In [23]:
```

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C:\Users\wamul\OneDrive\Desktop\Computer-Logic-and-Programming\untitled8.py

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```python
# -*- coding: utf-8 -*-
"""
Created on Thu Mar 12 10:05:19 2026

@author: wamul
2.  Fibonacci sequence is a sequence of natural order. The sequence is:
1, 1, 2, 3, 5, 8 etc. where it is a series of numbers that may start with 0 and 1, and
each subsequent number is the sum of the two preceding numbers. For this exercise, start with 1.

Use a for loop compute and display first 20 numbers in the sequence.


"""
first = 1
second = 1

print("Fibonacci Sequence (First 20 Numbers):")

for i in range(1, 21):
    print(first)
    next_number = first + second
    first = second
    second = next_number
```

Console 1/A

```
In [23]: %runfile C:/Users/wamul/OneDrive/Desktop/Computer-Logic-and-
Programming/untitled8.py --wdir
Fibonacci Sequence (First 20 Numbers):
1
1
2
3
5
8
13
21
34
55
89
144
233
377
610
987
1597
2584
4181
6765
```

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C:\Users\wamul\OneDrive\Desktop\Computer-Logic-and-Programming\untitled9.py

untitled7.py* | untitled8.py* | untitled9.py* | untitled10.py* | untitled11.py*

```python
# -*- coding: utf-8 -*-
"""
Created on Thu Mar 12 10:06:53 2026

@author: wamul
3.  Create a text file that contains employee last name and salary. Read in this data.
Determine the bonus rate based on the chart below. Use that rate to compute bonus.
For each line display the employee's last name, salary and bonus.
After the loop display the sum of all bonuses paid out.

"""
file = open("employees.txt", "r")

total_bonus = 0

print("Employee Bonus Report")

while True:
    lastname = file.readline().strip()
    if not lastname:
        break

    salary = float(file.readline().strip())

    if salary >= 100000:
        rate = 0.20
    elif salary >= 50000:
        rate = 0.15
    else:
        rate = 0.10

    bonus = salary * rate
    total_bonus += bonus

    print(f"{lastname}  Salary: ${salary:,.2f}  Bonus: ${bonus:,.2f}")

file.close()

print(f"\nTotal Bonuses Paid: ${total_bonus:,.2f}")
```

Console 1/A

```
In [25]: %runfile C:/Users/wamul/OneDrive/Desktop/Computer-Logic-and-
Programming/untitled9.py --wdir
Employee Bonus Report
George  Salary: $50,000.00  Bonus: $7,500.00
Wamz  Salary: $75,000.00  Bonus: $11,250.00
Smith  Salary: $45,000.00  Bonus: $4,500.00
Johnson  Salary: $120,000.00  Bonus: $24,000.00
Wamulume  Salary: $60,000.00  Bonus: $9,000.00

Total Bonuses Paid: $56,250.00

In [26]:
```

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C:\Users\wamul\OneDrive\Desktop\Computer-Logic-and-Programming\untitled10.py

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```python
# -*- coding: utf-8 -*-
"""
Created on Thu Mar 12 10:09:21 2026

@author: wamul
4.  Create a text file with item, quantity and price. Read through the file one line at a time.
Compute the extended price (quantity x price). For each line display the item, quantity, price and exten
After the loop display the sum of all the extended prices, the count of the number of orders and the ave
"""
file = open("items.txt", "r")

total_extended = 0
count = 0

print("Order Report")

while True:
    item = file.readline().strip()
    if not item:
        break

    quantity = float(file.readline().strip())
    price = float(file.readline().strip())

    extended = quantity * price

    total_extended += extended
    count += 1

    print(f"{item}  Qty: {quantity}  Price: ${price:.2f}  Extended: ${extended:.2f}")

file.close()

average = total_extended / count

print(f"\nTotal Sales: ${total_extended:,.2f}")
print("Number of Orders:", count)
print(f"Average Order: ${average:,.2f}")
```

Console 1/A

```
Johnson  Salary: $120,000.00  Bonus: $24,000.00
Wamulume  Salary: $60,000.00  Bonus: $9,000.00

Total Bonuses Paid: $56,250.00

In [26]: %runfile C:/Users/wamul/OneDrive/Desktop/Computer-Logic-and-
Programming/untitled10.py --wdir
Order Report
Widget  Qty: 10.0  Price: $50.00  Extended: $500.00
Hammer  Qty: 2.0  Price: $10.00  Extended: $20.00
Saw  Qty: 4.0  Price: $8.00  Extended: $32.00
Drill  Qty: 3.0  Price: $100.00  Extended: $300.00
Wrench  Qty: 5.0  Price: $20.00  Extended: $100.00

Total Sales: $952.00
Number of Orders: 5
Average Order: $190.40

In [27]:
```

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C:\Users\wamul\OneDrive\Desktop\Computer-Logic-and-Programming\untitled11.py

untitled7.py* untitled8.py* untitled9.py* untitled10.py* untitled11.py*

```python
@author: wamul
5.  Create a text file with student last name, district code (I or O) and number of credits taken.
Compute tuition owed (credits taken x cost per credit).
Cost per credit for in district students (district code I) is 250.00.
Out of district students pay 500.00 per credit. For each line display student last name, credits taken
After the loop display sum of all tuition owed and the number of students.
"""

file = open("students.txt", "r")

total_tuition = 0
count = 0

print("Tuition Report")

while True:
    lastname = file.readline().strip()
    if not lastname:
        break

    district = file.readline().strip()
    credits = float(file.readline().strip())

    if district == "I":
        cost_per_credit = 250.00
    else:
        cost_per_credit = 500.00

    tuition = credits * cost_per_credit

    total_tuition += tuition
    count += 1

    print(f"{lastname}  Credits: {credits}  Tuition: ${tuition:,.2f}")

file.close()

print(f"\nTotal Tuition Collected: ${total_tuition:,.2f}")
print("Number of Students:", count)
```

Console 1/A

```
In [28]: %runfile C:/Users/wamul/OneDrive/Desktop/Computer-Logic-and-
Programming/untitled11.py --wdir
Tuition Report
Lina  Credits: 12.0  Tuition: $3,000.00
Lisa  Credits: 10.0  Tuition: $2,500.00
Nthemba  Credits: 12.0  Tuition: $6,000.00
George  Credits: 16.0  Tuition: $8,000.00
Elizabeth  Credits: 15.0  Tuition: $3,750.00

Total Tuition Collected: $23,250.00
Number of Students: 5

In [29]:
```